

1. We consider the following linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, 2, \dots, n, \quad (1)$$

where Y_i is the dependent variable, X_i is the explanatory variable, and u_i is the error term. In practice, β_0 , β_1 , and the population regression line $\beta_0 + \beta_1 X$ are unknown. Therefore, we must use data to estimate these unknown values.

The Ordinary Least Squares (OLS) estimator finds the estimated regression line as close as possible to the observed data, where closeness is measured by the sum of squared errors:

- Here, given some values of b_0 and b_1 (given certain intercept and slope), under the model in (1), the error in predicting Y_i given X_i can be written as: $Y_i - (b_0 + b_1 X_i)$, for each i (i from 1 to n).
- Therefore, the sum of squared errors can be written as:

$$\sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2 = [Y_1 - (b_0 + b_1 X_1)]^2 + \dots + [Y_n - (b_0 + b_1 X_n)]^2.$$

The OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$, which are the estimators for the true values β_0 and β_1 , are obtained by solving the following minimization problem:

$$\min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2$$

- Intuitively, we search over all possible values of b_0 and b_1 , and find the values that minimize the sum of squared errors, in practice we solve the first order conditions of the minimization problem.

2. Refer to lecture notes and supplementary material.

3. First, we notice that

$$\begin{aligned}
& \sum_{i=1}^n (X_i - \bar{X})^2 \\
&= \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) \\
&= \sum_{i=1}^n X_i^2 - \sum_{i=1}^n 2X_i\bar{X} + \sum_{i=1}^n \bar{X}^2 \\
&= \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + n\bar{X}^2 \\
&= \sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 \\
&= \sum_{i=1}^n X_i^2 - n\bar{X}^2.
\end{aligned} \tag{2}$$

Second, we have

$$\begin{aligned}
& \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) \\
&= \sum_{i=1}^n (X_i Y_i - X_i \bar{Y} - \bar{X} Y_i + \bar{X} \bar{Y}) \\
&= \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n X_i \bar{Y} - \sum_{i=1}^n \bar{X} Y_i + \sum_{i=1}^n \bar{X} \bar{Y} \\
&= \sum_{i=1}^n X_i Y_i - n\bar{X} \bar{Y} - n\bar{X} \bar{Y} + n\bar{X} \bar{Y} \\
&= \sum_{i=1}^n X_i Y_i - n\bar{X} \bar{Y}.
\end{aligned} \tag{3}$$

Combining (2) and (3), we obtain

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n X_i Y_i - n\bar{X} \bar{Y}}{\sum_{i=1}^n X_i^2 - n\bar{X}^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}. \tag{4}$$

4. (a) Given that the classroom has 25 students, then the OLS regression's prediction is:

$$\widehat{TestScore} = 640.3 - 4.93 \times 25 = 517.05. \tag{5}$$

That is, the OLS regression's prediction for that classroom's average test score is 517.05.

(b) For the case with 21 students (last year),

$$\widehat{TestScore} = 640.3 - 4.93 \times 21 = 536.77. \tag{6}$$

For the case with 24 students (this year),

$$\widehat{TestScore} = 640.3 - 4.93 \times 24 = 521.98. \quad (7)$$

Therefore, the OLS regression's prediction for the change in the classroom's average test score is $\Delta \widehat{TestScore} = 521.98 - 536.77 = -14.79$, where $\Delta TestScore$ denotes the change in the classroom average test score.

Alternatively, according to the interpretation of $\hat{\beta}_1$, we have

$$\Delta \widehat{TestScore} = \hat{\beta}_1 \times \Delta CS = -4.93 \times (24 - 21) = -14.79, \quad (8)$$

where ΔCS denotes the change in the class size.

(c) According to the formulas of the OLS estimators,

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}, \quad (9)$$

which implies that

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}. \quad (10)$$

This implies that for the current example, where the sample average of class size is 22.8, $\hat{\beta}_0 = 640.3$ and $\hat{\beta}_1 = -4.93$, the sample average of the test scores is equal to $640.3 - 4.93 \times 22.8 = 527.896$.